

SHIVARAJ Y

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CAREER OBJECTIVE

Motivated Electronics and Communication Engineering undergraduate with a strong foundation in VLSI design, digital electronics, Verilog, and semiconductor concepts. Seeking an entry-level opportunity in Physical Design / RTL Verification where I can apply my technical skills, contribute to innovative semiconductor projects, and enhance my expertise in chip design and verification.

TECHNICAL SKILLS

Core: Digital Electronics, CMOS Fundamentals, ASIC Design Flow.

HDL: Verilog.

EDA Tools: Cadence Virtuoso.

Programming Languages: C, Python, TCL.

SOFT SKILLS

Quick Learner , Problem-Solving , Teamwork , Adaptability.

EDUCATION

B.E in Electronics and Communication Engineering	2023 – 2027
<i>Gopalan College of Engineering and Management</i>	<i>Average CGPA: 7.67</i>
Pre-University Course in PCMC	2021 – 2023
<i>New Sri Chaitanya P.U. College</i>	<i>Percentage: 77%</i>
Secondary School (SSLC)	2020 – 2021
<i>Vidya Residential English Medium High School</i>	<i>Percentage: 87%</i>

TECHNICAL TRAINING

VLSI Physical Design Training – VLSIMINDS , A Division of S2C Semiconductors Pvt. Ltd., Bengaluru	Jul 2026
– Trained on the complete ASIC Physical Design flow using Cadence EDA tools. – Gained hands-on experience in Floorplanning, Power Planning, Placement, CTS, Routing, and Static Timing Analysis (STA). – Practiced DRC/LVS verification, congestion analysis, and timing closure. – Developed practical knowledge of TCL scripting and Linux commands for EDA tool automation.	

PROJECTS

Smart Automated Ration Dispensing Machine | *Major Project (Ongoing)*

- Developing an automated ration dispensing system for accurate and efficient distribution.
- Integrating sensors, embedded hardware, and automation to minimize manual effort.

GPRS Controller with 16 SRAM Macros – RTL-to-GDSII Physical Design

- Implemented the complete RTL-to-GDSII Physical Design flow for a GPRS Controller integrating 16 SRAM macros and 20,198 standard cells.
- Performed floorplanning, macro placement, power planning, placement, CTS, routing, and physical-design closure using Cadence Innovus.
- Optimized macro placement, halos, routing channels, blockages, power connectivity, and standard-cell placement to reduce congestion and improve routability.
- Performed STA and timing optimization for a target frequency of 2 MHz, analyzing setup/hold WNS, TNS, critical paths, clock latency, and skew.
- Performed physical signoff including DRC/LVS, IR-drop, power analysis, and post-route timing verification, followed by final GDSII generation.

VLSI Design - SRAM memory RTL to GDS Flow

- Implemented SRAM memory design flow from RTL coding to GDSII generation using ASIC design methodologies.
- Performed RTL design, synthesis, floor planning, placement, clock tree synthesis, routing, and physical verification.
- Worked on Verilog-based RTL development and analyzed timing, area, and power optimization.

IoT-Based Smart Portable Cold Box | *Mini Project*

- Designed a portable cold storage system with IoT-based temperature monitoring.
- Integrated temperature sensing and real-time monitoring using embedded systems.

INTERNSHIP

Trainer – Io Brain Innovative Labz

2026 – Present

- Conduct weekly one class for 6th-grade students.
- Teach basic robotics, electronics, Arduino, and STEM concepts.
- Conduct simple hands-on activities and demonstrations.

PUBLICATIONS

Design and Implementation of an IoT-Enabled Thermoelectric Cold Box Powered by Hybrid Charger

- Published in *Swanirman Sunirmit Publications of Research (SSPR)*, ISSN: 2583-2654, Volume 6, Issue 1, March 2026.

ACHIEVEMENTS

- Qualified for **1st Round** of the **ISRO Robotics Challenge**, organized by the Indian Space Research Organisation.
- Won **2nd Prize** in **TECH NEXUS 2025 – Project Expo**, organized by Dr. T. Thimmaiah Institute of Technology.
- Won **2nd Place** in **GETOCS V6.0 2026 – Paper Presentation and RC Car Race**, organized by Gopalan College of Engineering and Management.

EXTRA-CURRICULAR ACTIVITIES

- Served as a Volunteer and Organizer for **WRO India 2026 (World Robot Olympiad India)**, assisting in event management, participant coordination, and competition logistics.
- Participated in **Yuktikalpa - 2025 Tech Expo**, organized by BMS College of Engineering.
- Participated in **AIKYA - 2025 Project Expo**, organized by East Point College of Engineering and Technology.
- Participated in **Smart India Hackathon (SIH)** at the college-level selection for 2 consecutive years.
- Co-ordinator of the **Institution's Innovation Council (IIC)**, actively involved in organizing innovation, technical, and entrepreneurship programs.